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STATS 101C: Introduction to Statistical Models and Data Mining

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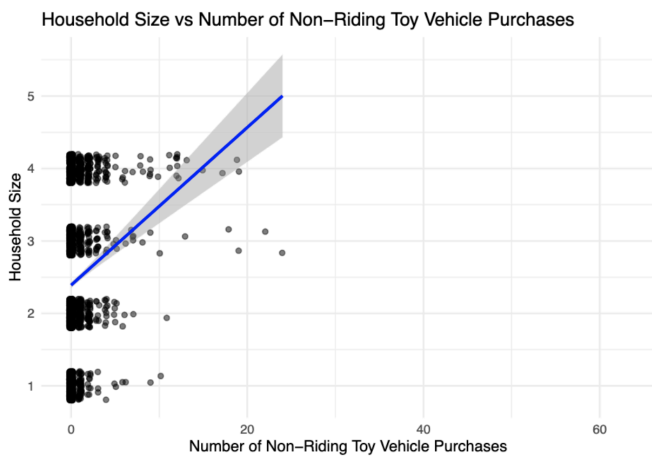
Household Size Regression Model Report

Introduction

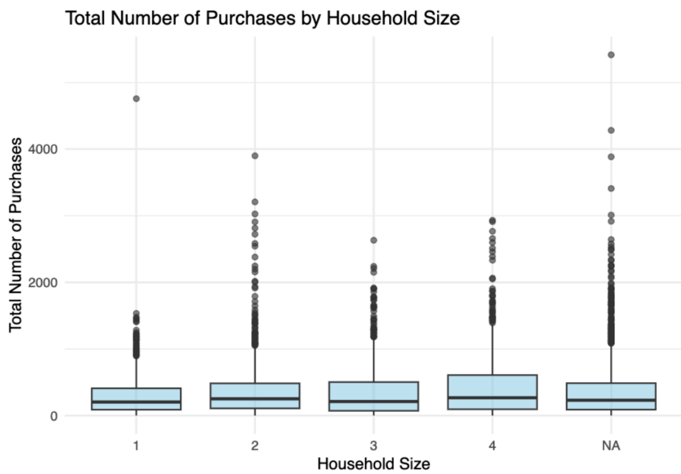
Given the task of estimating household size based on Amazon purchase history data and survey responses, our group believed certain categories may indicate a purchase made for a family. For example, anything related to babies, children's toys, or motherhood likely indicates children in the household, meaning a larger household size. Outside of categories, we also logically assumed users with more purchases and more total money spent are likely buying items for more than just themselves, again pointing to a higher household size. Of course these assumptions will not always hold as many people buy children-related items as gifts for friends or families, and people with more disposable income in general will likely buy more items. Purchasing behavior has a wide variety depending on the individual and our models had to rely on lots of assumptions despite this variation. Still, overall, we assumed these would be the most promising variables to look at. We did not use any outside sources.

Exploratory Data Analysis

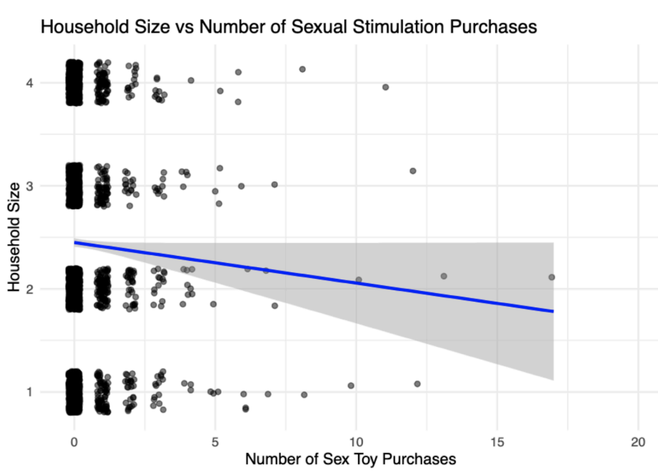


There is an extremely strong positive correlation between the number of non-riding toy vehicle purchases and household size. A toy such as this is likely only consumed by children, so it can strongly indicate a larger household size.

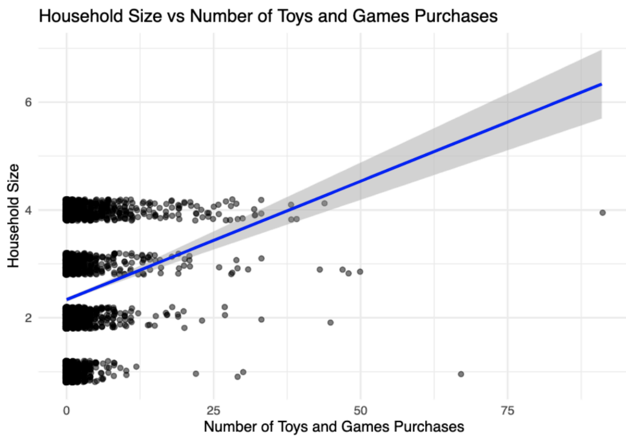
Similarly to non-riding toy vehicles, teether purchases also indicate larger household sizes. Teethers are used by babies and indicate that a household has children and therefore the prediction for household size would be large.



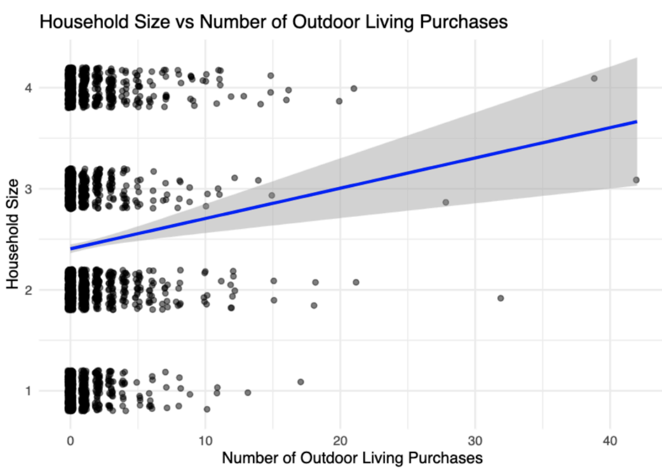
All household sizes complete the same number of total purchases on average. Households of size two are generally the accounts with the largest number of total purchases, though this is not a dramatic difference.



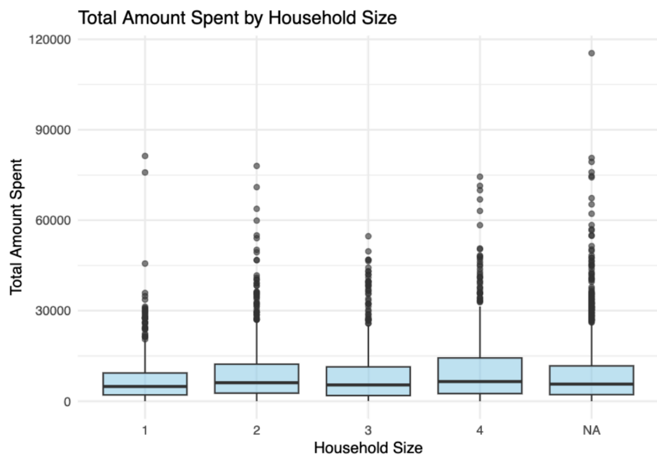
The number of sexual stimulation category purchases appears to be an indicator of households of one or two, as childless couples are more likely to buy this sort of item.



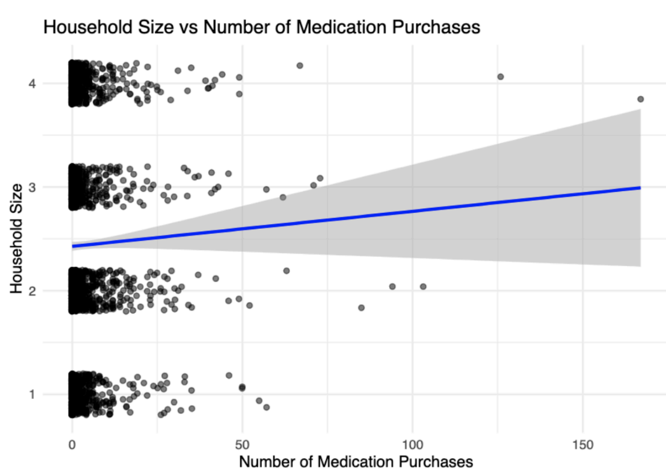
As expected, an increased number of purchases of toys and games indicates larger household size. These types of purchases suggest that there are children present in the household.



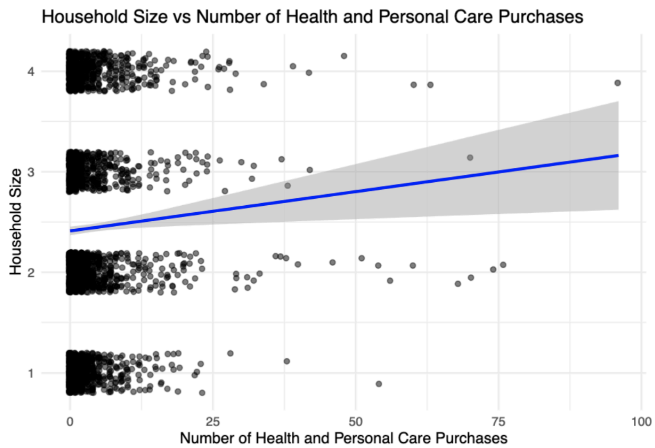
The number of outdoor living purchases tends to correlate to a larger household size. This might be because people with children are more likely to have a larger yard or that people with children are more likely to do outdoor activities than households of one or two.



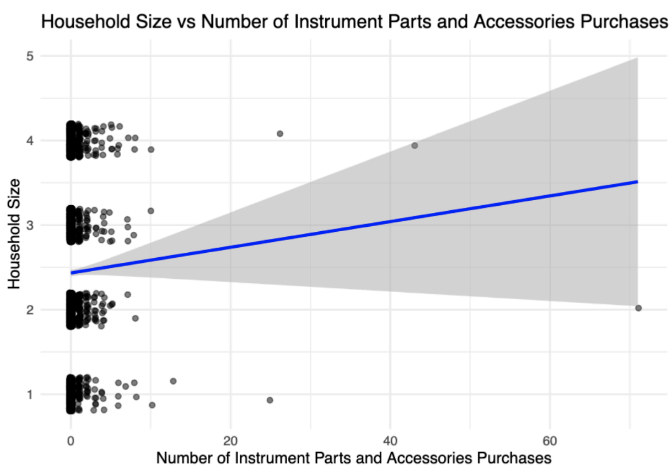
The total amount spent does not seem to be entirely indicative of household size. This graph shows that households of size three actually spend the least amount of money on Amazon, while households of one spend the most, in terms of maximums. In terms of medians, all household sizes appear to spend comparable amounts indicating a lack of a correlation or pattern.



An increased number of medication purchases appears to slightly correlate to a larger household size. Because medication is used by adults and children indiscriminately, an increased use of it might generally suggest more inhabitants in a household. This is not a strong correlation, however, probably because the use of medication between individuals is not universally uniform.



There is a similar trend in health and personal care purchases to household size as was observed in medication purchases. These are similar commodities.



The graph demonstrates that with a purchase of instrument parts and accessories there may be a slight increase in the prediction of the household size. However this graph has a particularly large confidence interval.

Preprocessing

To preprocess the data, first we had to combine the amazon purchase history data with the train/test split data set with a full join. Because there are multiple entries per user ID in the amazon purchase history, we then had to reshape the data to be one row per user. For example, new per-user columns would have averages of predictors like price of item for each user and new predictors like total number orders. This was achieved primarily through functions in the dplyr package. From these first changes we were able to have the combined information of the survey response and amazon history summarized to a row per household instead of a row per purchase. Next we converted the text in the responses to factors to make it easier to code our models. Then, as suggested by our teaching assistant Jay, we found the top 50 product categories and one-hot encoded the categories to turn the columns to binary outputs instead. Lastly, we ensured we removed any variables that were not included in the testing data set such as age, race, and income as we wanted to ensure our data would not rely on things that were not available when testing our model on the test data.

We later changed the top 50 to top 200 to see if that would improve the model quality (it generally did not). To consolidate the categorical information, we next grouped similar relevant categories together. We created several new binary columns that identified if a user had ordered something from a toy/game related top 200 category, a baby related top 200 category, or a pet related top 200 category. This was accomplished with string detection in the column names with the string package and the rowSums function; each user received a 1 in each respective new column if the rowSums was greater than or equal to 1 for the selected columns, or 0 otherwise. We kept the categories of incontinence protection and pajamas since those were important in the original top 200 category model, but removed all other individual category columns.

In our candidate model/model evaluation section, we will discuss models that utilized data from several steps of the preprocessing process as we believe grouping the categories differently was relevant to test in different models.

Candidate Models/ Model Evaluation/ Tuning

All of the models we will discuss in this section are some type of regression tree (except for the xgboost gradient boosted decision tree forest). We had tried a couple other types of models (ex basic multiple linear regression and smoothing spline), but anything unrelated to a regression tree was far worse than any type of regression tree. Thus, for the purposes of this project, we will focus on different variations of regression trees (and we will mention the xgboost model, which was surprisingly bad). We also tried some other versions of regression trees, but these ones made the most sense to discuss for this project.

Table 1: Model Explanation

Model	**Variables used**	**Hyperparameters**
Model 1: Regression Tree (1)	50 categorical predictors Survey responses Total: 57	Rpart default No tuning
Model 2: Regression Tree (2)	200 categorical predictors Survey response Total: approx 257	Rpart default No tuning
Model 4: Regression Tree (3)	Combined categorical	Rpart default

	predictors Survey response Total: 14	
Model 3: Tuned Regression Tree (1)	200 categorical predictors Survey response Total: approx 257	Maxdepth = 1 Minsplit = 59
Model 5: Tuned Regression Tree (2)	Combined categorical predictors Survey response Total: 14	Maxdepth = 2 Minsplit = 7
Model 6: Xgboost	50 categorical predictors Survey responses Total: 57	Objective = reg:squarederror Eta = 0.1 Max_depth = 6 Subsample = 0.8 Colsample_bytree = 0.8 Nrounds = 300 Verbose = 1

Table 2: Model Performance in Kaggle

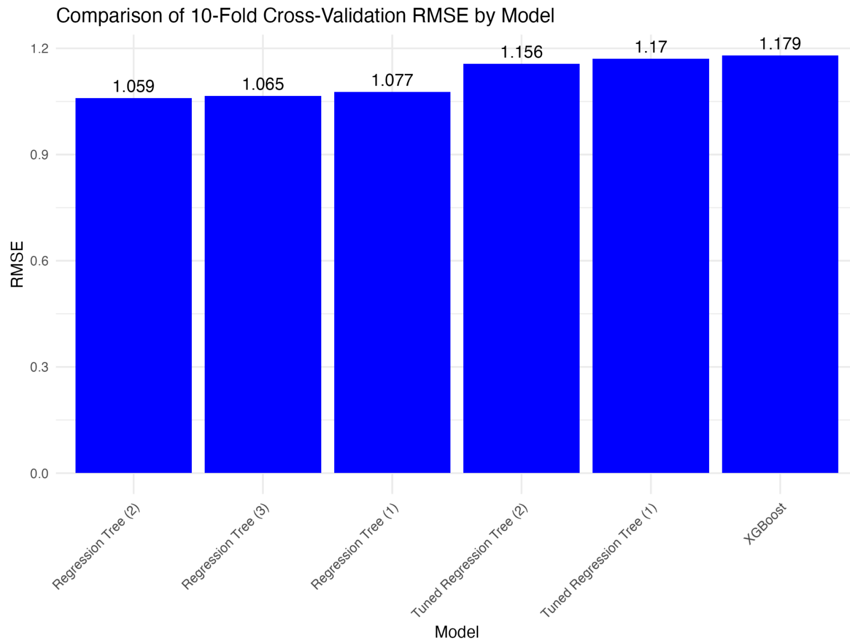
Model	**Kaggle RMSE**

Model 1: Regression Tree (1)	1.08832
Model 2: Regression Tree (2)	1.10222
Model 4: Regression Tree (3)	1.10693
Model 3: Tuned Regression Tree (1)	1.09463
Model 5: Tuned Regression Tree (2)	1.09253
Model 6: Xgboost	1.51095

Table 3: Model Performance through Cross Validation

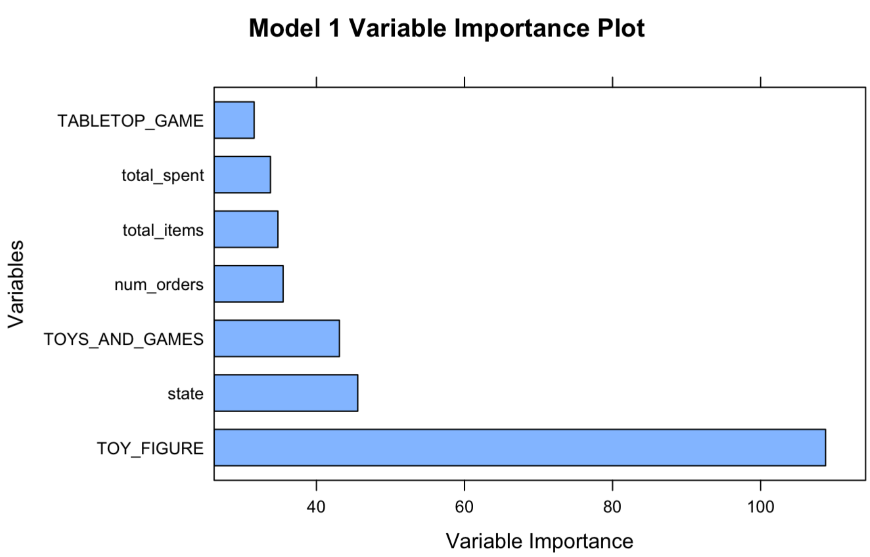
Model	**10 Fold Cross Validation Performance – RMSE**
Model 1: Regression Tree (1)	1.076517
Model 2: Regression Tree (2)	1.058935
Model 4: Regression Tree (3)	1.065372
Model 3: Tuned Regression Tree (1)	1.170383
Model 5: Tuned Regression Tree (2)	1.15607
Model 6: Xgboost	1.179344

Figure: Comparing Model Performance by Cross Validation

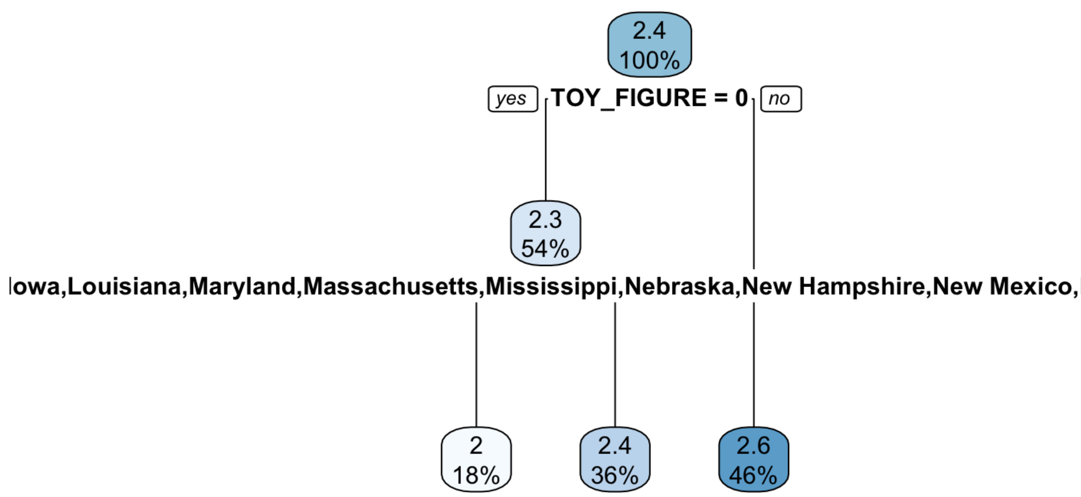


Model 1: Regression Tree (1) - includes top 50 categorical predictors

This model is a fairly simple regression tree created using rpart. The predictors include the total amount of money spent, the total number of items purchased, the total number of amazon orders, the average price, the user gender and state, and whether or not the user purchased something in each of the 50 most common categories.



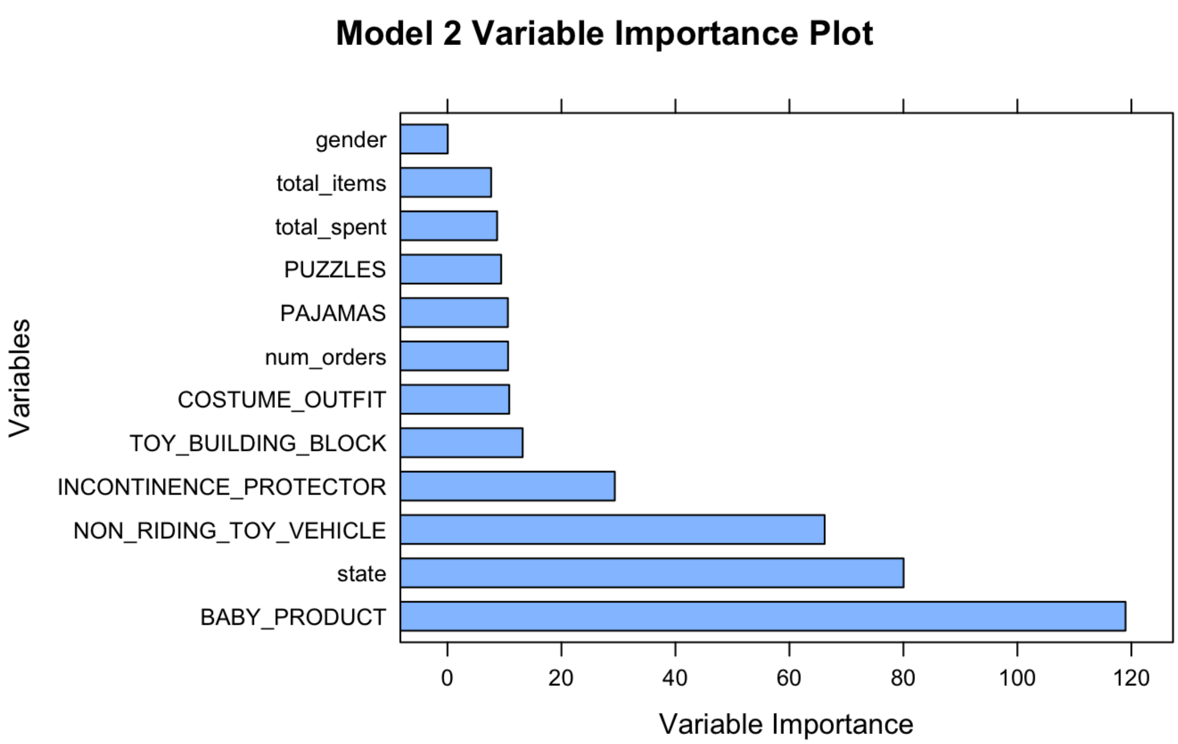
In this model, whether someone purchased something in the "Toy Figure" category is by far the most important variable. The next most important variables are the state someone lives in, whether they bought something from the "Toys and Games" category, the number of orders they've placed, the total items placed, the total amount spent, and whether they purchased something in the "Tabletop Game" category respectively.



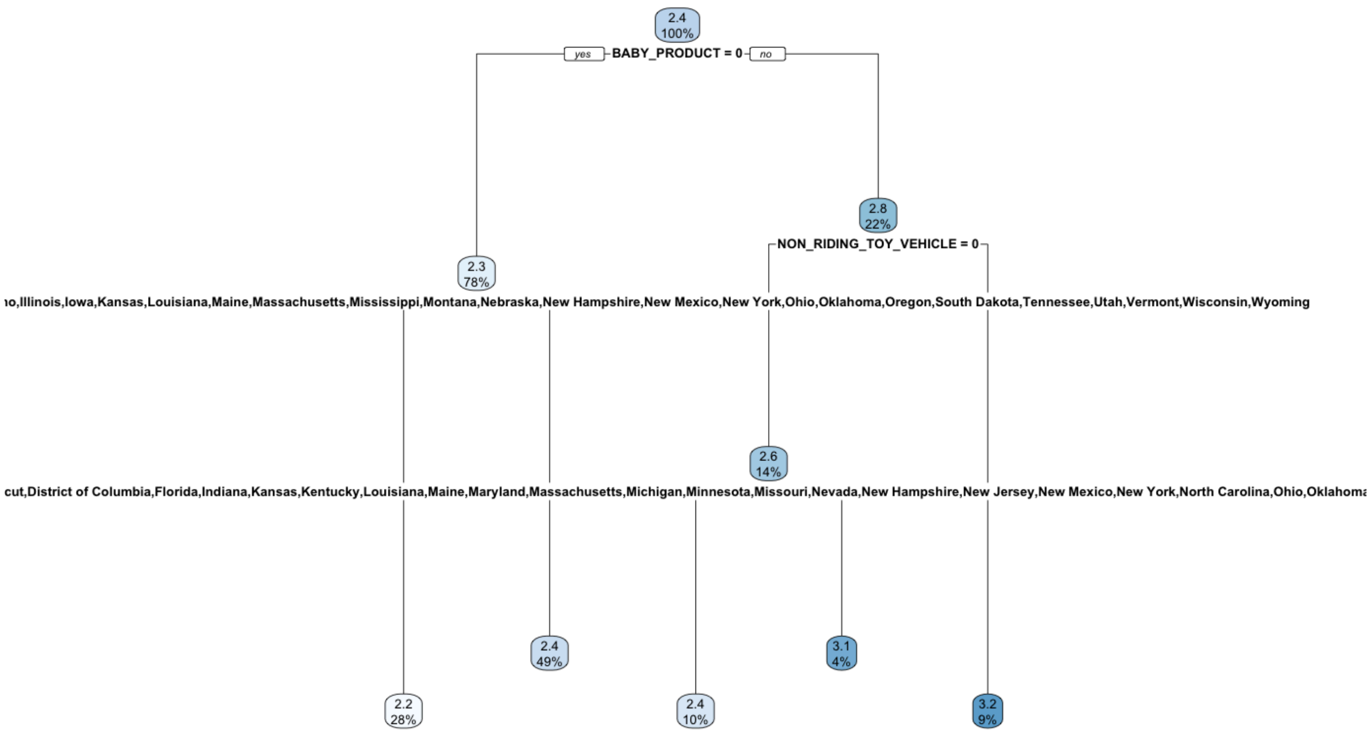
In this model, if someone did not buy anything considered a "Toy Figure", their household size is estimated to be 2.6 people. But, if they did not buy anything in that category, the state they live in matters for the prediction. For example, if someone did not buy a "Toy Figure" and they live in Alabama, Alaska, Arizona, Arkansas, Colorado, Delaware, District of Columbia, Idaho, Iowa, Louisiana, Maryland, Massachusetts, Mississippi, Nebraska, New Hampshire, New Mexico, New York, Ohio, Oklahoma, Oregon, South Carolina, South Dakota, Tennessee, Utah, or Vermont, their household size is predicted to be 2. If they did not buy a "Toy Figure" and they live in any other state, their household size is predicted to be 2.4. In subsequent models, we will continue to treat states separately rather than group by region as there does not seem to be a relationship between US state regions and which side of that split a state was placed on. However, we will explore different ways to treat the categories of purchases.

Model 2: Regression Tree (2) - includes top 200 categorical predictors

This model is very similar to the last model. The only difference is that it uses the top 200 most common categories. The other variables are still the total amount of money spent, the total number of items purchased, the total number of amazon orders, the average price, and the user gender and state. We wanted to test models with different numbers of categories to get a gauge of how many top categories we needed. According to our 10 fold cross validation, including the top 200 categorical predictors has a slightly smaller root mean squared error than the model with the top 50 categorical predictors, however according to the Kaggle scores, the regression tree with the top 50 categorical predictors is slightly better.



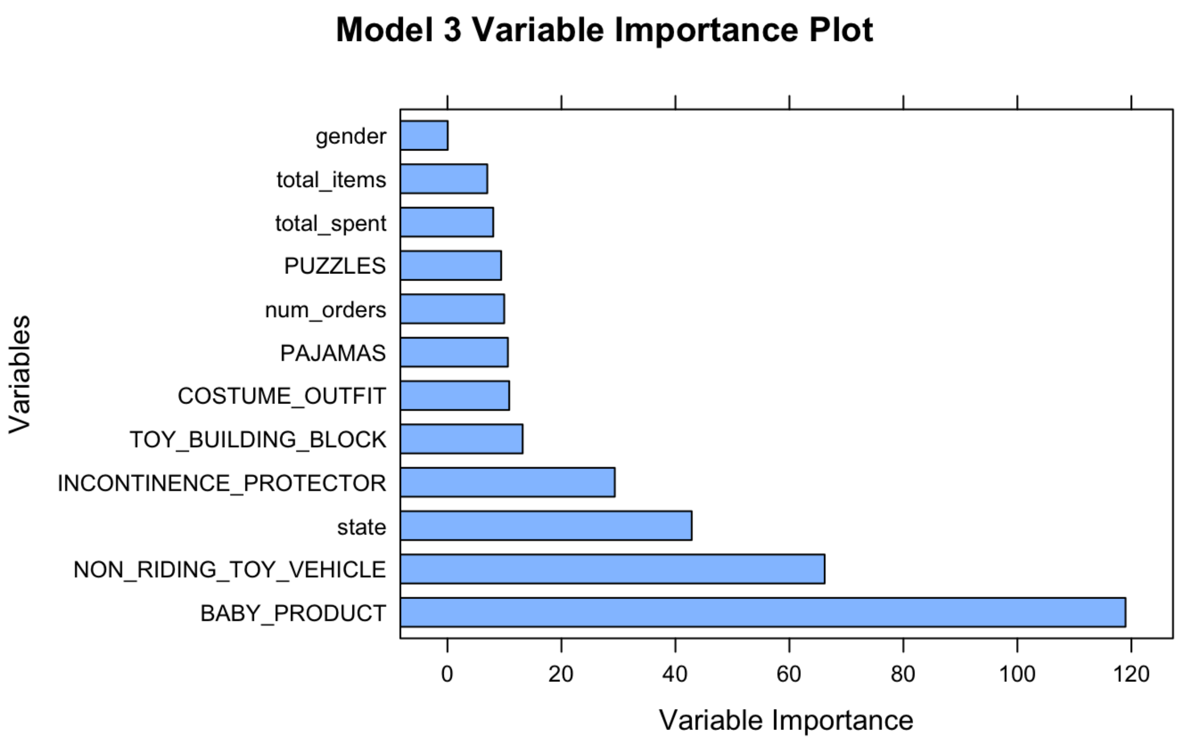
In this model, the most important predictor by far is whether someone bought something considered a “Baby Product”. The next most important predictor is again state, followed by the categories “Non-Riding Toy Vehicle”, “Incontinence Protector”, “Toy Building Block”, and “Costume Outfit”, the number of orders placed, whether someone purchased pajamas, whether someone purchased puzzles, the total amount spent, the total items, and finally the gender. The first several predictors are far more important than the last few.



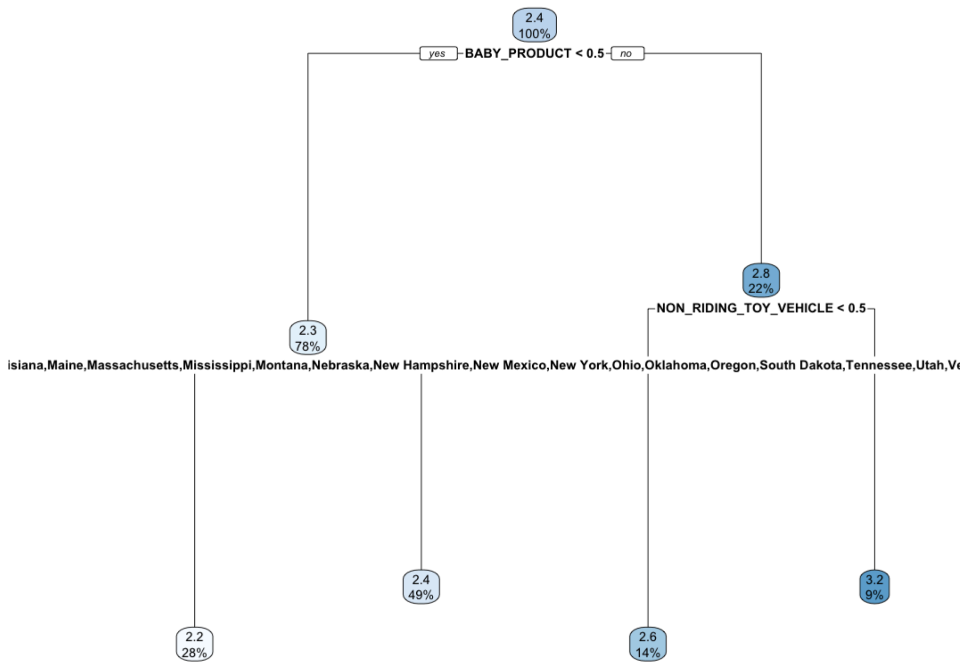
This model is slightly more complicated than model one, so I will just discuss a few prediction paths here, though all are in the regression tree figure. If someone bought a “Baby Product” and a “Non-Riding Toy Vehicle”, the household size is estimated to be 3.2. If they did not buy a “Baby Product”, they are predicted to have a household size of about 2.2 if they live in Alabama, Arizona, Arkansas, Colorado, Delaware, Idaho, Illinois, Iowa ,Kansas, Louisiana, Maine, Massachusetts, Mississippi, Montana, Nebraska, New Hampshire, New Mexico, New York ,Ohio, Oklahoma, Oregon, South Dakota, Tennessee, Utah, Vermont, Wisconsin, or Wyoming, but a household size of about 2.4 if they live in any other state.

Model 3: Tuned Regression Tree (2)

This model is quite similar to model 2, however we added hyperparameter tuning. We figured since including the top 200 predictors shows some promise (based on the cross validation performed in R), it was worth it to tune the model with the top 200 predictors. This surprisingly ended up being one of our worst models, both according to cross validation in R and the Kaggle RMSE.



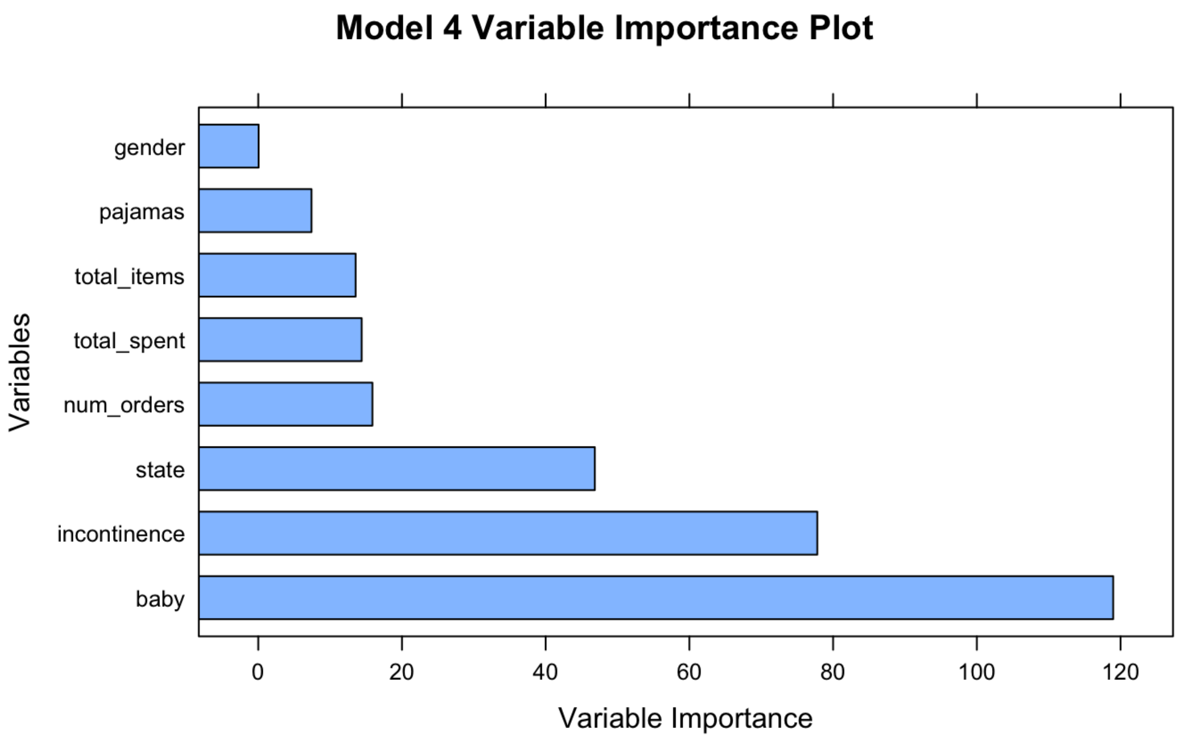
In this model, “Baby Product” was still the most important category, followed by “Non-Riding Toy Vehicle”, and state. There were several more potentially important variables, as listed in the output, though not important enough to make it onto the regression tree.



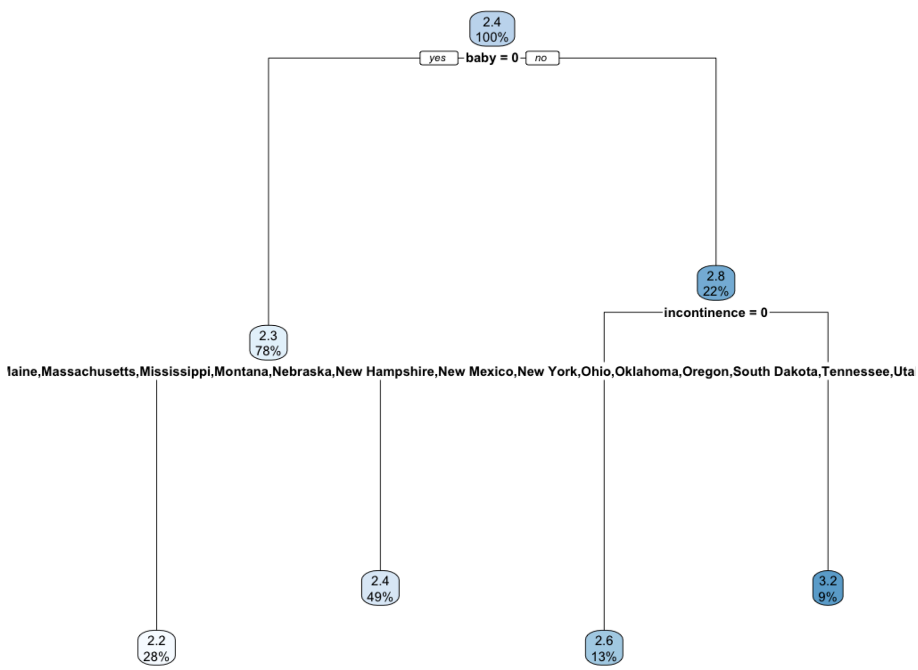
Tuning the model did not actually change the model in this case. For instance, if someone bought a “Baby Product” and a “Non-Riding Toy Vehicle”, the household size is still estimated to be 3.2. This makes sense since the root mean squared error for the tuned model is actually a bit worse than model 2, indicating that changing the model in model 2 (through changing the maximum depth or minimum split of the model) will not improve on model 2. Thus, there would be no reason for this tuned model to have different predictions than model 2.

Model 4: Regression Tree (3) - includes combined categorical predictors

This model still keeps the same numerical predictors (the total amount of money spent, the total number of items purchased, the total number of amazon orders, the average price) as well as state and gender, but combines similar categories from the top 200 categories that may be relevant to household size. We figured this would be a good way to make the model simpler while still including the most important information. The combined categories include anything related to toys, babies, or pets, and we also kept the singular categories for incontinence protection and pajamas because those were also some of the most important categories in other models.



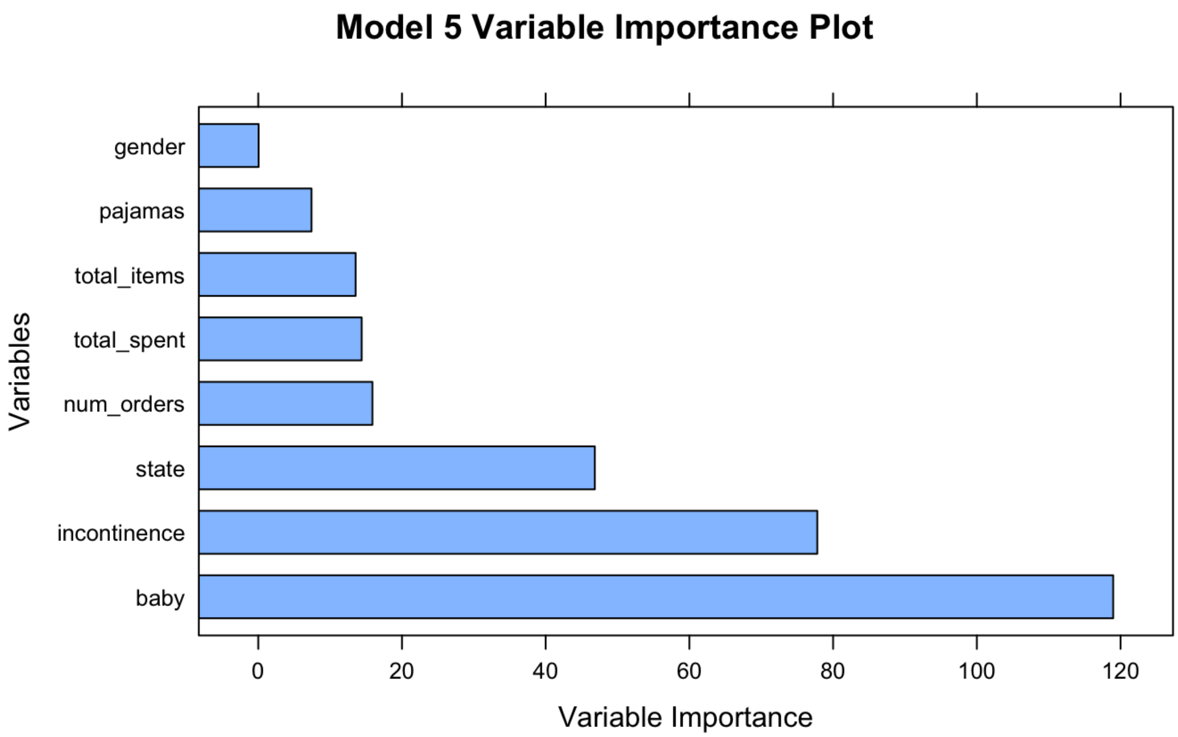
Now whether someone purchased anything from a baby related category is the most important predictor. The next most important predictor is the incontinence protection category, which, though not in any of the past models, had relatively high variable importance in many of the previous models. Though not the first thing someone may think of as a predictor of household size, this makes sense, since, unfortunately for many new mothers, giving birth can cause incontinence. Thus, incontinence products could indicate someone in the house gave birth recently, indicating a larger household size. Furthermore, incontinence products are far less likely to be given as a gift than toys or baby products, so it makes sense that this factor, though not only caused by birth, would be closely linked to household size. State is once again an important enough variable to be featured in the regression tree. There are again several more potentially important predictors, though not important enough to make it into the regression tree.



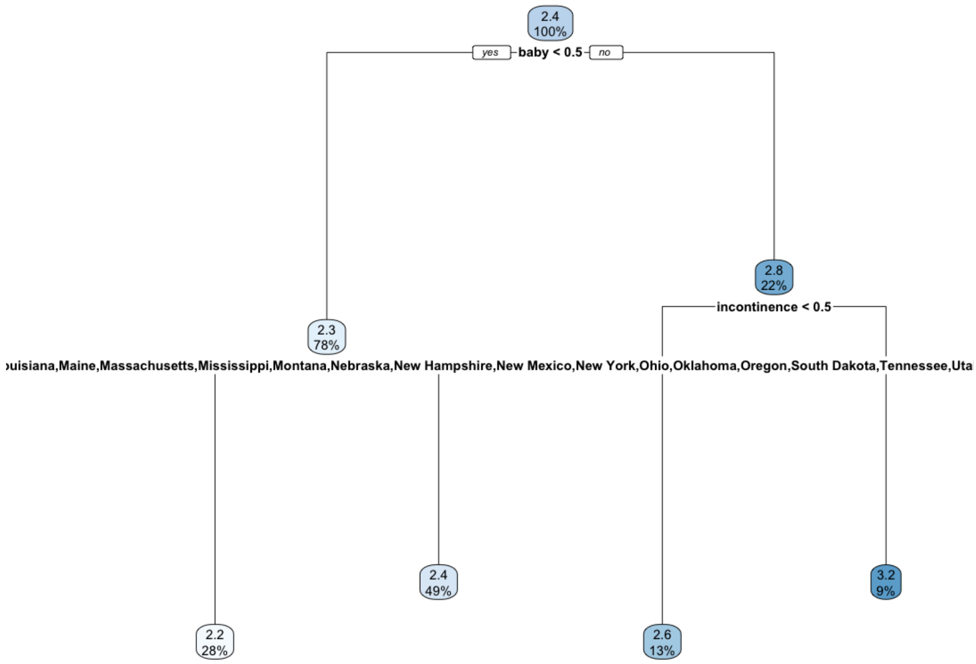
In this model, if someone bought a baby related product and also bought incontinence protectors, their household size is predicted to be 3.2. This is one of the highest predictions of any node in any model we've seen, but it makes sense this combination would indicate a larger household size as this logically seems to imply someone in the user's household has given birth and is caring for at least one baby. If they bought a baby product but not any incontinence protectors, they are predicted to have a household size of 2.6, which is the next highest terminal node in this model. If they have not purchased a baby product and live in Alabama, Arizona, Arkansas, Colorado, Delaware, Idaho, Illinois, Iowa ,Kansas, Louisiana, Maine, Massachusetts, Mississippi, Montana, Nebraska, New Hampshire, New Mexico, New York ,Ohio, Oklahoma, Oregon, South Dakota, Tennessee, Utah, Vermont, Wisconsin, or Wyoming, the user is predicted to have a household size of 2.2, which is the smallest predicted household size of a terminal node in this model.

Model 5: Tuned Regression Tree (2)

We created this model to see if we could tune hyperparameters to improve on the regression tree with combined categorical predictors. This still uses the predictors the total amount of money spent, the total number of items purchased, the total number of amazon orders, the average price per item, user state, user gender, and the same combined categories from model 4. This model however did not improve on model 4 and was actually a bit worse.

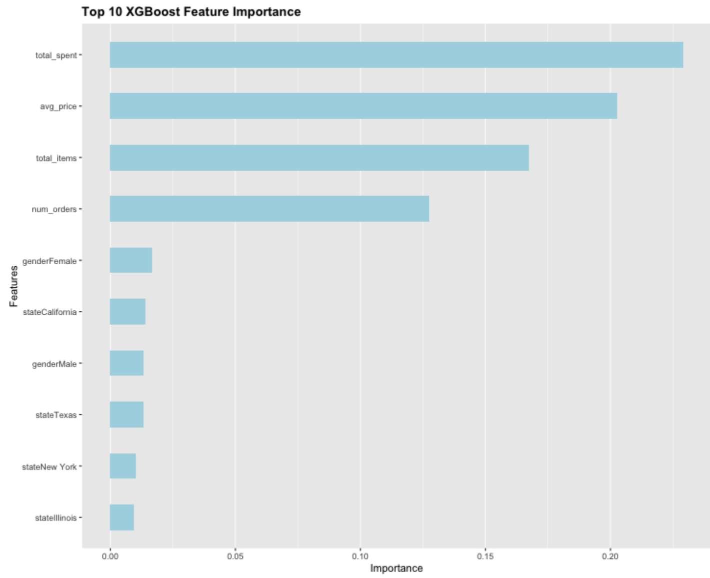


Purchase of a baby product, purchase of an incontinence protector, and user's state are again the most important predictors. The actual regression tree is essentially the same in this model as in model 4. This is for the same reason models 2 and 3 are the same.



Once again, the tuning did not find hyperparameters that produced a better model, so the model outputted is the same as the default rpart makes without tuning. It is still the case that if a user bought a baby product but not any incontinence protectors, they are predicted to have a household size of 2.6. If they have not purchased a baby product and live in Alabama, Arizona, Arkansas, Colorado, Delaware, Idaho, Illinois, Iowa ,Kansas, Louisiana, Maine, Massachusetts, Mississippi, Montana, Nebraska, New Hampshire, New Mexico, New York ,Ohio, Oklahoma, Oregon, South Dakota, Tennessee, Utah, Vermont, Wisconsin, or Wyoming, the user is predicted to have a household size of 2.2.

Model 6: Xgboost



This model relies on gradient boosted decision trees to try to predict the household size based on survey responses and amazon purchases. An xgboost model is built sequentially with each focusing on reducing the residuals from the previous model. To incorporate the amazon history the model continues the strategy of including the top 50 product categories as a binary category and feeds the model information on whether the user purchased something within these categories.

The most important features in the xgboost model included the total spent, average price, and total items respectively. Although this model did surprisingly lack efficiency and accuracy in predicting, the model was able to demonstrate important information about the impacts of the variables. For example, if a user has a low total spending their prediction in household size would sharply drop. Additionally it was found that if the user identified with the gender male the prediction would increase in comparison to being female.

This model ultimately attempted to try a different strategy than the differing variations of regression trees. This model failed the most and demonstrated that, in this case, sticking to a more basic regression model would be the best option in terms of the end goal: minimizing the RMSE while predicting household size.

Discussion of Final Model

All of the models discussed (except the xgboost model) have fairly similar RMSE scores, so it is difficult to choose a best model. The three different (non-tuned) regression tree models discussed in the candidate model section all were tested in Kaggle and reported a RMSE score within 0.02 of each other. The two tuned regression trees' RMSE scores were within 0.003 of each other in Kaggle. The cross validation results are slightly different than the Kaggle results, which makes sense, but the models perform so similarly that it matters if you use Kaggle or cross validation to determine which was the "best" model. According to cross validation, model 2 was the best, but according to Kaggle, model 1 was the best. Notably, the tuned models perform slightly better than their respective counterparts based on the Kaggle even though they do not perform better with cross validation. This makes sense, as hyperparameter tuning is designed to improve the model performance on unseen data. Given this information, some of these models may have outperformed each other due to chance. Ultimately it can be concluded that some type of regression tree was the best model in this project of predicting household size, but we cannot confidently say which of our models will perform best on the unseen data that will determine the final Kaggle leaderboard rankings.

For the purposes of this assignment, we will discuss model 1 as our "final model" since it technically scored highest on Kaggle. This model has the benefit of being the simplest of the candidate models discussed in this report, so it is unlikely to overfit to the training data and should work well on unseen data (as it did on the first half of the released Kaggle data). Retaining only the top 50 categories is beneficial because it allows the model to focus on what are likely to be the most important categorical predictors. However, there still may be important categories that are not particularly popular overall, but are purchased almost exclusively by parents with many children (who would thus have a larger household size). If such a category existed, it would be very valuable to this regression task, but would not be captured by this model.

This model could also benefit from more demographic information about the users which could interact with the categorical purchase history information. For example, if we knew the age of each user, the model may be able to distinguish more easily between people buying toy figures for their own children instead of as gifts. Logically, if the user is over a certain age, say 60, it is unlikely they are purchasing toys for their own children, so children related items may be gifts for grandchildren. Additionally, though incontinence protection products are not present in model 1, age may be especially valuable to models 2-5 since age is helpful to gauge whether this product likely is being purchased due to pregnancy/birth related incontinence issues or age related incontinence issues. Both women who've recently given birth and older individuals are more prone to incontinence issues, but older individuals are more likely to have adult children who no longer live with them, and therefore likely have a smaller household size than the women who have recently given birth.

In general, regression trees are typically praised for their ability to create interpretable models, and deal with non linear data with diverse aspects. However, it is important to note that regression trees should be used with caution as they sometimes overfit, have high variance, and can lack the simple ability to capture linear trends. In this report, a regression tree performed the best because of the ability to handle the large data set combining both survey responses and the amazon purchase history. It is noted that the limitations of this model likely include not including basic trends such as survey response demographic data and could have over-fitted in terms of the provided train data set, which would then pick up noise and pattern that could not be successfully applied to the testing data. This is an issue for any statistical model however, so it does not discredit the logic behind building model 1, or any of our regression trees for that matter.

Given the difficulty of the regression task with the provided data sets, it can be concluded that our model would likely perform better given more time to preprocess and clean the dataset. With the combining of the amazon purchase history and the survey data there was lots of categorical data which made it difficult to identify and choose which variables to include and yet most importantly how. If given more time, we may have been able to explore interaction terms between different purchase categories, thus improving the model. For example, if someone purchased a baby product and incontinence protectors, they may be more likely to have small children, and thus a larger household size, than people who purchased just one of those items. Combining this with added demographic information would be particularly powerful for future models.

Appendix: Team Member Contributions

Eden:

Helped Elly with data cleaning/processing (though Elly did the bulk of that step in the end). Explored and created several models, including models 1 through 5 discussed in this report, as well as a basic linear regression model, a smoothing spline, and a logistic tuned regression tree, all of which performed very poorly and were therefore not reported on. Wrote report introduction. Wrote report information for models 1 through 5 and created respective variable importance plots and tree plots. Conducted 10-fold cross validation on models 1 through 5. Contributed significantly to the discussion of the final model. Communicated consistently in the group chat. Helped edit report.

Elly:

Performed the data cleaning and processing and shared code with other group members. Continued to explain the process and reasoning in the report paper. Contribute to creating candidate models that were ultimately submitted to the kaggle competition but lacked to find any progress in comparison to Eden's model. Created necessary tables and graphs for the report which compared the models performances. Explained my own xgboost model that was 1 of the 6 candidate models discussed in the report. Consistently communicated in the group chat with other group members to help and encourage collaboration. Created a shared google docs folder with all necessary information including a spreadsheet I created which tracked the progress of our reports in addition to who was completing which part.

Zane:

Performed exploratory data analysis on the data. Modified and manipulated the data sets in order to extract more meaningful visualizations.

Alice and Val:

Both Alice and Valérie focused on the regression project. At the original group meeting helped discuss ideas for cleaning data, candidate models, and report formatting. Kept up to date with group communication and Kaggle submissions. Ran code to ensure reproducibility and contributed to editing this report.